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Q1) Write program demonstrates how to use regular expressions in Python to match and search for patterns in text.

Programiz
Python Online Compiler

main.py

Visualize

Run

```
import re

text = "Python programming is easy"

a = re.match("Python", text)
b = re.search("easy", text)

if a:
    print("Match found:", a.group())
else:
    print("No match found")

if b:
    print("Search found:", b.group())
else:
    print("No search found")
```

Output

```
Match found: Python
Search found: easy
```

Q2) Implement a basic finite state automaton that recognizes a specific language or pattern. In this example, we'll create a simple automaton to match strings ending with 'ab' using python.

main.py

Visualize

Run

```python
def fsa(s):
    state = 0

    for ch in s:
        if ch == 'a':
            state = 1
        elif ch == 'b':
            if state == 1:
                state = 2
            else:
                state = 0
        else:
            return False

    return state == 2

s = input("Enter a string containing a and b: ")

if fsa(s):
    print("String Accepted")
else:
    print("String Rejected")
```

Output

```
Enter a string containing a and b: abab
String Accepted
```

Q3) Write program demonstrates how to perform morphological analysis using the NLTK library in Python.

main.py

Visualize

Run

```
word = input("Enter a word: ")

if word.endswith("ies"):
    base = word[:-3] + "y"
elif word.endswith("ing"):
    base = word[:-3]
elif word.endswith("ed"):
    base = word[:-2]
elif word.endswith("s") and not word.endswith("ss"):
    base = word[:-1]
else:
    base = word

print("Base form:", base)
```

Output

```
Enter a word: cities
Base form: city
```

Q4) Implement a finite-state machine for morphological parsing. In this example, we'll create a simple machine to generate plural forms of English nouns using python.

main.py

Visualize

Run

```python
def plural(word):
    if word.endswith(("s", "sh", "ch", "x", "z")):
        return word + "es"
    elif word.endswith("y") and len(word) > 1 and word[-2] not in "aeiou":
        return word[:-1] + "ies"
    else:
        return word + "s"

word = input("Enter a singular noun: ")
print("Plural form:", plural(word))
```

Output

```
Enter a singular noun: city
Plural form: cities
```

Q5) Use the Porter Stemmer algorithm to perform word stemming on a list of words using python libraries.

main.py

Visualize

Run

```python
def stem(word):
    if word.endswith("ies"):
        return word[:-3] + "i"
    elif word.endswith("ing"):
        return word[:-3]
    elif word.endswith("ed"):
        return word[:-2]
    elif word.endswith("s"):
        return word[:-1]
    else:
        return word

words = ["playing", "played", "plays", "studies", "running"]

for word in words:
    print(word, "->", stem(word))
```

Output

```
playing -> play
played -> play
plays -> play
studies -> studi
running -> runn
```